Old Exam Questions
ML Supervised Techniques

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How this file is organized

This file is separate from the course summary and is meant as a clean practice set. Exact duplicates were removed. When essentially the same question appeared in multiple old exams, that is marked with *Asked multiple times.* inside the question box. The wording was lightly normalized where the scans were messy, but the mathematical content and answer choices were kept intact.

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Introduction & Basics

Question 1: Area Under Curve

Consider the following classification results of a binary classifier tested on 8 samples:

$\bar{g}(x)$	y
0.1	1
−0.2	−1
2.0	1
−0.8	−1
−1.2	−1
1.5	1
−0.3	1
0.9	−1

Compute the area under the ROC curve (AUC). Give the answer rounded to two decimal places.

Answer: _____

Question 2: Evaluation Measures

Assume a binary classification test set with 120 samples where 36 are true negatives and 16 are false negatives. There are twice as many total positives as total negatives in the test set.

Compute accuracy and balanced accuracy. Give answers rounded to two decimal places.

Accuracy = _____ Balanced accuracy = _____

Question 3: Bias–Variance Decomposition

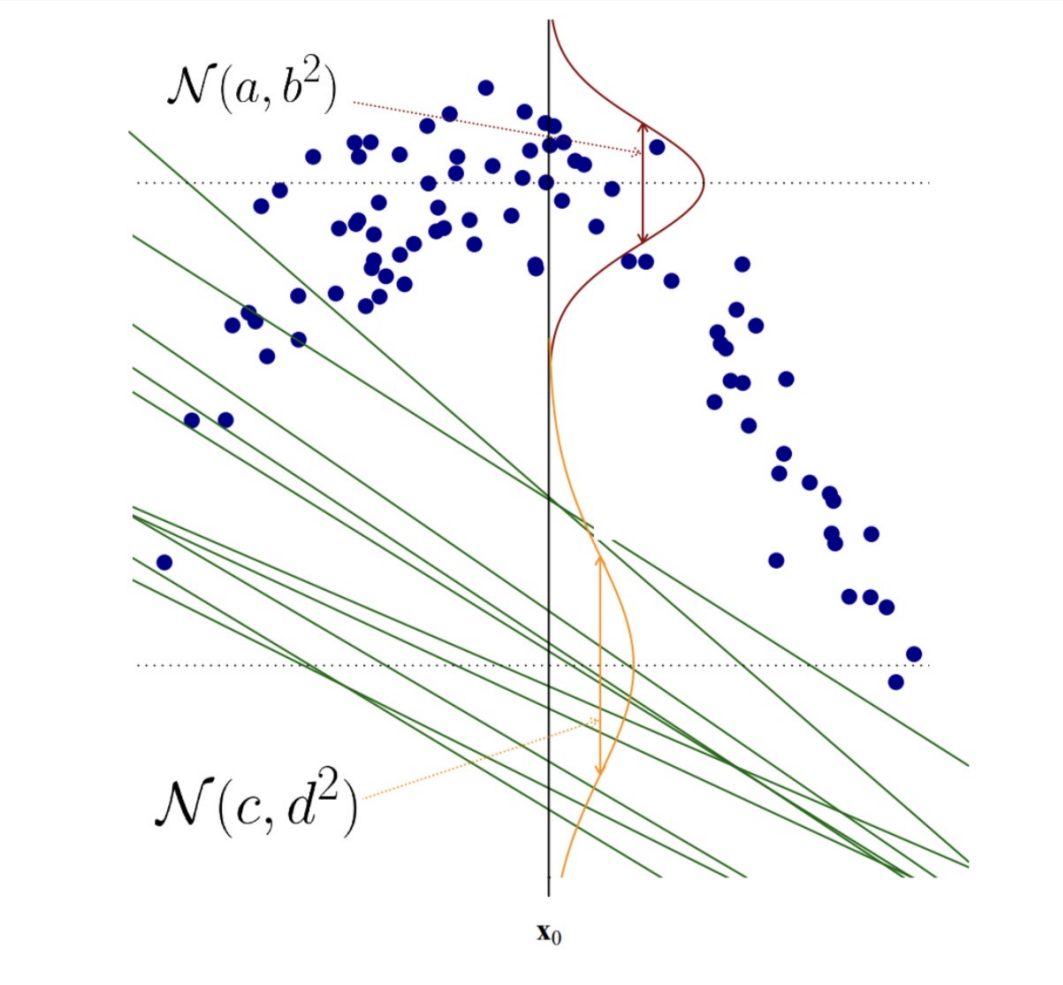
Assume that the prediction of a model is described by the distribution:

$$g(X) \sim \mathcal{N}(a, b^2).$$

The distribution of the true values is:

$$y \sim \mathcal{N}(c, d^2).$$

The distributions are shown in the plot below. Set $a = 2$, $b = 4$, $c = -3$, and $d = 6$.



Compute the following quantities. Give the answer as a real number rounded to two decimal places.

- Model variance: _____
- Unavoidable error: _____
- Bias: _____
- Expected prediction error: _____

Question 4: Medical Test Accuracy

In a group of 50 000 people, 45 000 are healthy (negative) and 5 000 are positive. A medical test has sensitivity 60% and specificity 80%. What is the accuracy of the test? Give the answer as a real number rounded to two decimal places.

Answer: _____

Question 5: Covid Test Probability

Assume currently 1% of the Austrian population have Covid (positive). Assume a certain Covid antigen test has 95% sensitivity and 90% specificity. A person says that the test shows positive. What is the probability that the person is healthy? Give the answer as a real number rounded to two decimal places.

Answer: _____

Question 6: Maximum Likelihood (Climate Model)

Assume you have a climate model with an important parameter θ. You are given five data points (the 10-year-average global temperatures for the past 50 years). With parameter choice θ_1, the probabilities to obtain the given data within a certain margin of error are:

$$(0.8,\ 0.9,\ 0.9,\ 0.7,\ 0.8).$$

With parameter choice θ_2, the probabilities are:

$$(0.9,\ 0.9,\ 0.5,\ 0.9,\ 0.9).$$

- Compute the likelihood for θ_1: _____
- Compute the likelihood for θ_2: _____
- Which parameter gives the better explanation? (Enter $i = 1$ or $i = 2$): _____

Question 7: Biased Coin: Rank Likelihoods

A biased coin has probability $\frac{1}{3}$ to show tails "T" and probability $\frac{2}{3}$ to show heads "H". Given the following sequences:

(1) HHHHHHTTTHH
(2) TT
(3) HTHTHT
(4) HHHH
(5) HTH

Rank them in descending likelihood according to the coin (most likely first). Fill in the indices:

$i_1 =$ _____ $i_2 =$ _____ $i_3 =$ _____ $i_4 =$ _____ $i_5 =$ _____

Question 8: Negative Log-Likelihood

Assume we flip a coin 4 times and observe the result $\{1, 1, 1, 0\}$ where head is 1.

Assume an estimator for p leads to a lower negative log-likelihood of the samples. Such an estimator is *better*. (True/False): _____

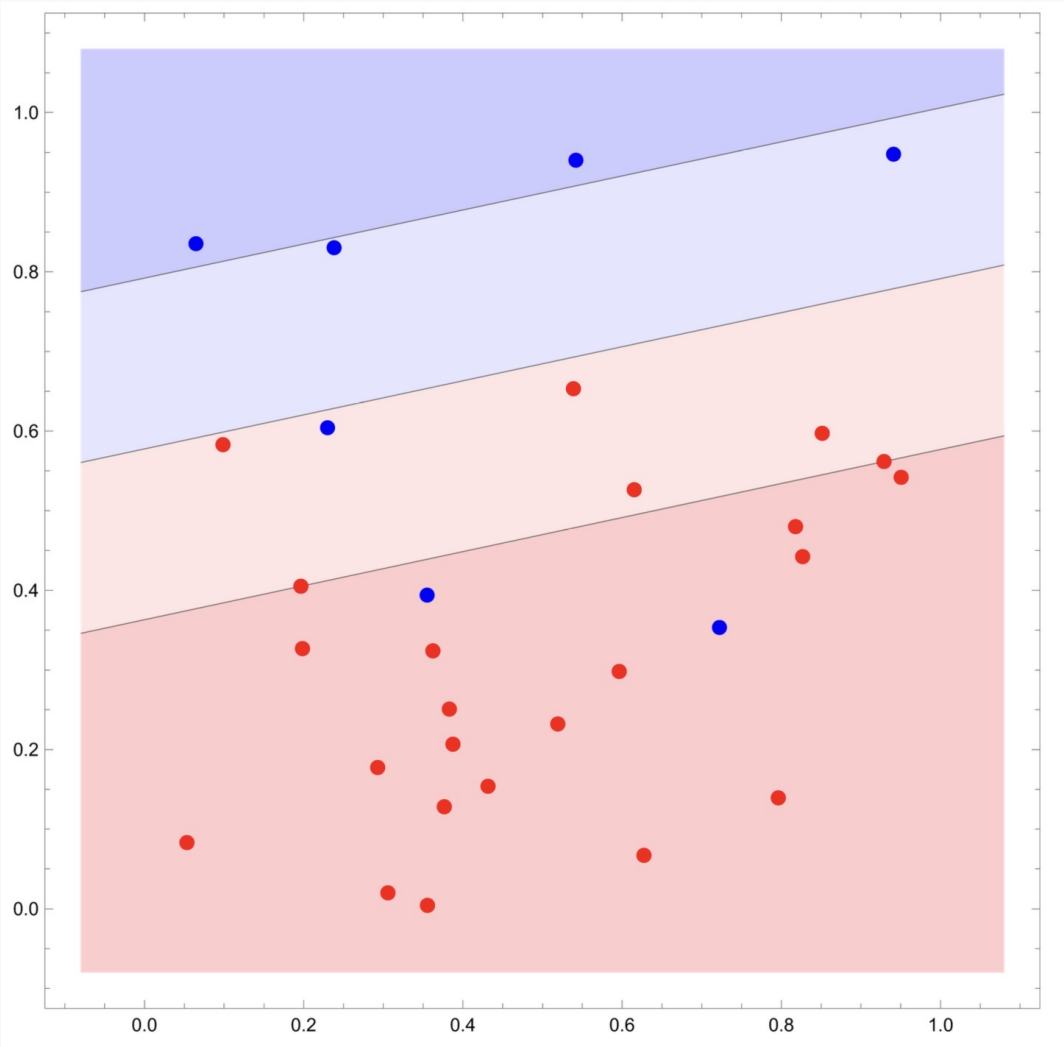
Compute the negative log-likelihood for:

- $p = 0.5$: _____
- $p = 0.58$: _____

Support Vector Machines

Question 9: Support Vectors

How many data points are support vectors?



Number of blue support vectors: _____

Number of red support vectors: _____

Question 10: Constrained Optimization

Consider the constrained optimization problem:

$$\min_{w_1,w_2} \frac{1}{2}(w_1^2 + 2w_2^2 + 7) \quad \text{s.t.} \quad w_1 + 6w_2 \geq 19.$$

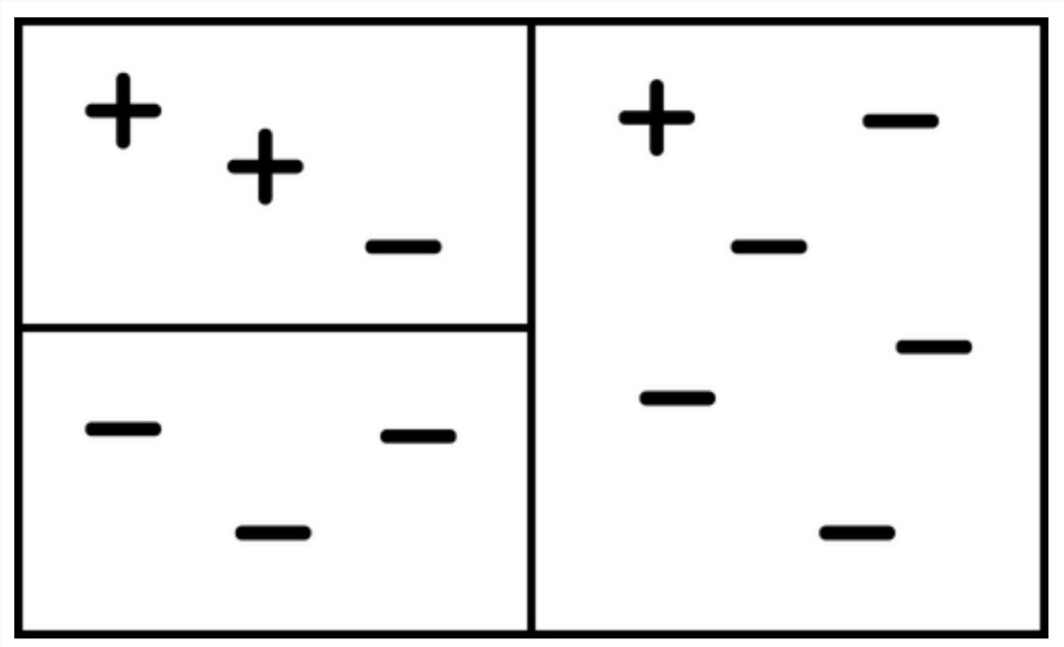
Compute the solution $(w_1^\star, w_2^\star)$ (integers).

$w_1^\star =$ _____ $w_2^\star =$ _____

Random Forests & Gradient Boosting

Question 11: Gini Impurity

Consider fitting a decision tree on binary data with classes "+" and "-". Assume that an intermediate iteration has resulted in the depicted partition.



What is the Gini impurity gain of the next optimal partition? Give the answer as a real number rounded to two decimal places.

Answer: _____

Question 12: Boosting: Sign of Strong Classifier

Given weak learners $g_1, \ldots, g_6$ and their weights $\alpha_1, \ldots, \alpha_6$. The output of the learners at a point x is:

$$g_1(x) = 1, \quad g_2(x) = -1, \quad g_3(x) = -1, \quad g_4(x) = -1, \quad g_5(x) = -1, \quad g_6(x) = -1.$$

The weights are:

$$\alpha_1 = 2.5,\ \alpha_2 = 0.5,\ \alpha_3 = 2.5,\ \alpha_4 = 1.5,\ \alpha_5 = 2.0,\ \alpha_6 = 0.5.$$

Compute the sign of $\sum_{m=1}^{6} \alpha_m g_m(x)$ at point x.

Answer: _____

Question 13: Decision Trees, Random Forests, and Boosting

Which statements about decision trees, random forests, and boosting are true? Multiple answers are possible.

- ☐ Information gain is a common splitting criterion for regression.
- ☐ A split maximizes the information gain if the produced subsets are completely homogeneous.

- ☐ Decision tree learning algorithms recursively split the training set into subsets.
- ☐ A common measure of variable importance in random forests is the mean accuracy decrease.
- ☐ In a random forest, each decision tree uses a random bootstrap subsample of the original data set.
- ☐ The Extreme regularized Boosting algorithm remains a state-of-the-art supervised learning method on tabular data.
- ☐ The maximal entropy of a data set with M classes is $\ln M$.
- ☐ In boosting, the main idea is that those samples that led to a small error (i.e. were classified well) in an iteration get a larger weight in the next iteration.

Question 14: AdaBoost

Select the correct statements about AdaBoost.

- ☐ In every step n, compute a misclassification error and classifier weight α_n. If classification for a data point was correct, leave corresponding weight unchanged. If classification was wrong, increase weight.
- ☐ Final output:
$$g(x) = \operatorname{sign}\left(\sum_{n=1}^{N} \alpha_n g_n(x)\right).$$
- ☐ Initialization: Weights are all initialized to the same value and sum up to 1.
- ☐ In every step, with given weights, find optimal classifier $g_n(x)$ to the training data.

Question 15: Ensemble Methods

Asked multiple times.

Tick the correct statements related to ensemble methods (several may be correct).

- ☐ To construct the out-of-bag random forest predictor corresponding to a specific data sample z, you only average trees which are trained on data that contain z.
- ☐ The bagging procedure increases the variance of a single model.
- ☐ Out-of-bag error of random forests is computed with the same samples that are used for fitting the model.
- ☐ In each iteration, AdaBoost decreases the weights of misclassified samples.
- ☐ Random forests use unpruned decision trees.
- ☐ In AdaBoost, in each iteration n, the weight α_n assigned to the classifier g_n depends on its classification error.
- ☐ In each AdaBoost iteration, the weak learner is trained on a weighted dataset.
- ☐ Boosting is a method to reduce bias.

Question 16: Ensemble Methods

Asked multiple times.

Tick the correct statements related to ensemble methods (several may be correct).

- ☐ Random forests use random subsets of features in each tree.
- ☐ AdaBoost uses random subsets of samples in each iteration.
- ☐ Bagging aims to decrease the bias of a single model.
- ☐ Boosting aims to decrease the bias of a single model.
- ☐ Bagging aims to decrease the variance of a single model.
- ☐ Ensemble methods are typically used to reduce variance.
- ☐ Bagging trains models sequentially (each model depends on previous).
- ☐ Random forests use bootstrap samples (sampling with replacement).

Question 17: Ensemble Methods

Asked multiple times.

Tick the correct statements related to ensemble methods (several may be correct).

- ☐ Bagging aims at reducing the bias of the ensemble model.
- ☐ Random forests use a random subset of features when splitting nodes.
- ☐ AdaBoost uses a random subset of features when splitting nodes.
- ☐ Boosting aims at reducing the bias of the ensemble model.
- ☐ Bagging trains models sequentially.
- ☐ Bagging aims at reducing the variance of the ensemble model.

Question 18: Boosting: Sign of Strong Classifier

Asked multiple times.

Given weak learners $g_1, \ldots, g_5$ and their weights $\alpha_1, \ldots, \alpha_5$. The output of the learners at a point x is:

$$g_1(x) = -1, \quad g_2(x) = -1, \quad g_3(x) = +1, \quad g_4(x) = +1, \quad g_5(x) = +1.$$

The weights are:

$$\alpha_1 = 5.0,\ \alpha_2 = 4.5,\ \alpha_3 = 6.5,\ \alpha_4 = 0.5,\ \alpha_5 = 1.0.$$

Compute the output of the strong classifier.

Answer: _____

Question 19: Boosting: Sign of Strong Classifier

Asked multiple times.

Given weak learners $g_1, \ldots, g_6$ and their weights $\alpha_1, \ldots, \alpha_6$. The output of the learners at a

point x is:

$$g_1(x) = -1,\ g_2(x) = +1,\ g_3(x) = +1,\ g_4(x) = -1,\ g_5(x) = +1,\ g_6(x) = +1.$$

The weights are:

$$\alpha_1 = 4.5,\ \alpha_2 = 0.5,\ \alpha_3 = 6.5,\ \alpha_4 = 4.5,\ \alpha_5 = 1.0,\ \alpha_6 = 0.5.$$

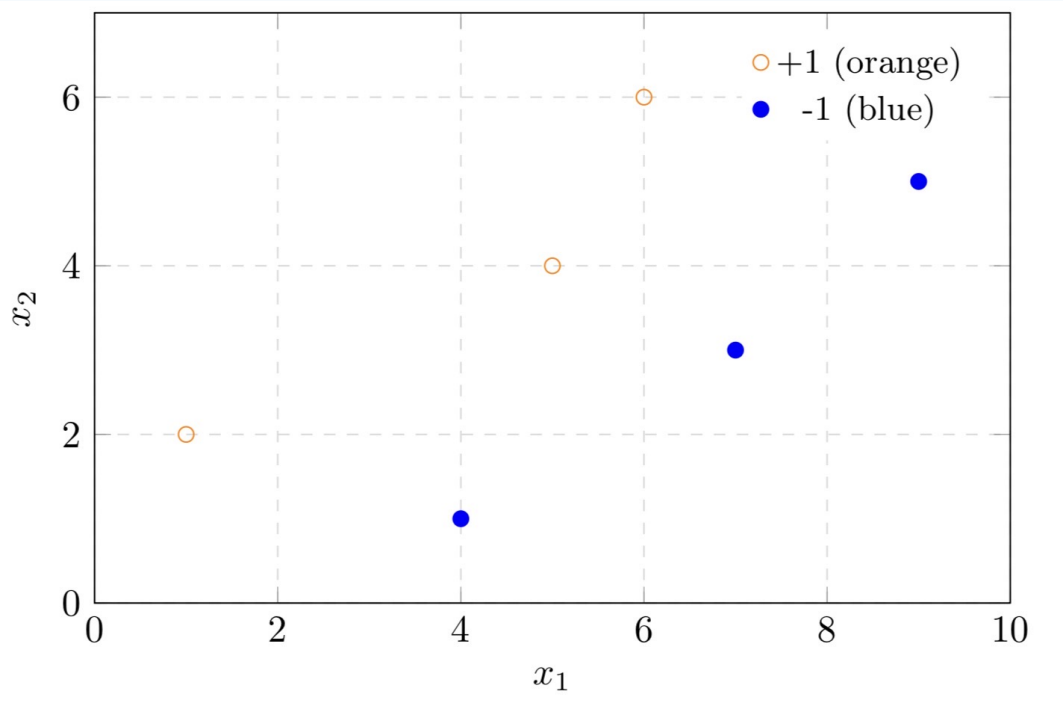
Compute the output of the strong classifier.

Answer: _____

Question 20: AdaBoost: Weight Increase

The plot above shows labeled points in 2D. Orange points have label $+1$, blue points have label -1. We use the AdaBoost algorithm with uniform weights at the beginning. Assume the first model is:

$$g_1(x) = \begin{cases} +1 & \text{if } x_2 < 3.5 \\ -1 & \text{else} \end{cases}$$



Which point(s) get weight increased due to incorporating g_1? Select one:

- ☐ $(5, 4)$
- ☐ $(7, 3)$
- ☐ $(6, 6)$
- ☐ $(9, 5)$
- ☐ $(4, 1)$
- ☐ $(1, 2)$

Logistic Regression

Question 21: Logistic Regression

Which of the following statements about Logistic Regression are correct? Select one or more:

- ☐ In Logistic regression, the probability distribution is given by:

$$p(y|x) = g(x) \text{ for } y = 1, \qquad p(y|x) = 1 - g(x) \text{ for } y = -1.$$

- ☐ In Logistic Regression, the discriminant function is not linear.
- ☐ The decision boundary of Logistic Regression is generally non-linear.
- ☐ Logistic Regression assumes that the outputs are Bernoulli distributed.
- ☐ The maximum likelihood solution for Logistic Regression cannot be computed in closed-form.
- ☐ The decision boundary of Logistic Regression for a binary classification task is given by $g(x) = 0.5$.
- ☐ In Logistic regression, the logistic function is given by $\sigma(z) = \left(1 \exp(z)\right)^{-1}$.
- ☐ Maximizing the negative log-likelihood for Gaussian noise is equivalent to minimizing the squared error.

Question 22: Logistic Regression

Asked multiple times.

Which of the following statements about Logistic Regression are correct? Select one or more:

- ☐ The logistic function $\sigma(z)$ is symmetric around the origin ($\sigma(z) = \sigma(-z)$).
- ☐ If $\|w\|$ is large, the classifier tends to output labels close to 0 or 1.
- ☐ The decision boundary is linear in x.
- ☐ The maximum likelihood estimate has a closed-form solution.
- ☐ The gradient of the negative log-likelihood is used in iterative optimization methods.
- ☐ Logistic Regression minimizes the squared error loss.
- ☐ Logistic Regression assumes Gaussian distributed noise.

Question 23: Logistic Regression

Asked multiple times.

Which of the following statements about Logistic Regression are correct? Select one or more:

- ☐ Logistic regression is a generative model.
- ☐ Logistic regression can only be used for linearly separable data.
- ☐ The sigmoid maps real numbers to the interval $(0, 1)$.
- ☐ The loss function used is convex.
- ☐ When training a Logistic Regression classifier one might get stuck in a local minimum of

the loss which may yield sub-optimal performance.

Question 24: Logistic Regression Output

Given $x = (1, 2, 3, 4, 5)^\top$ and $W = (1, -1, 1, -1, 1)^\top$. Compute the output of a Logistic Regression model (real-valued result; do not apply sign). Give the answer as a real number rounded to two decimal places.

Answer: _____

Question 25: Cross-Entropy Loss

Given two samples $(x_1, y_1) = (x_1, 1)$ and $(x_2, y_2) = (x_2, 0)$, and model outputs $g(x_1) = 0.9$ and $g(x_2) = 0.4$. Compute the cross-entropy loss (use natural logarithm). Give the answer rounded to two decimal places.

Answer: _____

Artificial Neural Networks

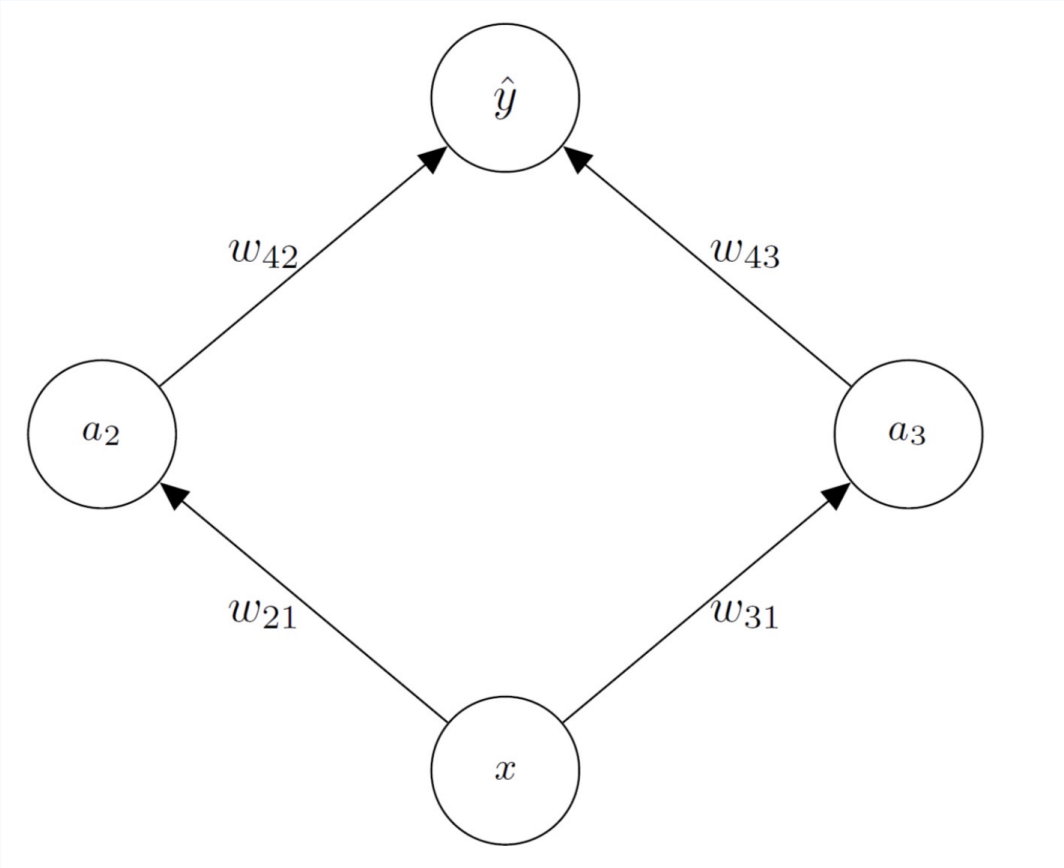
Question 26: Neural Networks

Which statements about neural networks are true? Multiple answers are possible.

- ☐ Vanishing gradient problem can be avoided by suitable loss functions.
- ☐ Let $J^{(l)}$ denote the Jacobian of layer l, where $l = 0$ is the input layer. Then we have the recursive formula: $\delta^{(l+1)\top}(t) = \delta^{(l)\top}(t)J^{(l)}$.
- ☐ Universal Approximation Theorem guarantees that neural networks can approximate any continuous function.
- ☐ Backpropagation is an instance of reverse mode automatic differentiation.
- ☐ Dropout and weight decay are regularization techniques that aim at avoiding underfitting.
- ☐ Let us denote the learning rate with $\eta > 0$, the weights with w, and the empirical risk with R. Then the gradient descent update rule reads: $w_{\text{new}} = w_{\text{old}} - \eta \nabla_w R$.
- ☐ A single-layer neural network cannot solve the OR classification problem.
- ☐ Universal Approximation Theorem guarantees that neural networks can approximate any continuous function.

Question 27: Backpropagation

Assume the following simple neural network with ReLU activation in the hidden layer and identity activation in the output layer:



The loss function is squared error: $L = \frac{1}{2}(y - \hat{y})^2$.

Given $x = 2$, $w_{21} = 2$, $w_{31} = 1$, $w_{42} = -1$, $w_{43} = 2$, $y = 1$, and learning rate $\eta = 0.01$, compute:

- $\hat{y}$: _____
- δ_2: _____
- $\frac{\partial L}{\partial w_{21}}$: _____
- w_{21}^{new}: _____
- w_{31}^{new}: _____

Special Network Architectures

Question 28: Sequence Models

Which statements are true? Multiple answers are possible.

- ☐ Transformer networks' computational demand scales exponentially with sequence length.
- ☐ We denote with R the recurrent weight matrix at time t, with f the activation function, and with $s(t-1)$ the preactivation at time $t-1$. For stable gradients, we need $\|R\,\mathrm{diag}(f(s(t-1)))\| \approx 1$.
- ☐ RNNs are Turing complete.

- ☐ The current hidden state depends on the current input and previous output.
- ☐ The attention mechanism is invariant to a simultaneous permutation of keys and values.
- ☐ In attention, queries and keys have the same feature dimension but different lengths.
- ☐ Sentiment classification of a text is a many-to-many task.
- ☐ LSTMs use an integrator to store information and gates to manage what information is stored in the integrator.

Question 29: Convolution

Consider the input matrix x:

$$x = \begin{pmatrix} 2 & 1 & 2 & 0 & 1 & 0 & 3 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 0 & 2 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 2 & 0 & 0 \\ 2 & 2 & 2 & 2 & 2 & 2 & 1 \\ 1 & 2 & 3 & 2 & 3 & 2 & 1 \end{pmatrix}$$

and the filter W:

$$W = \begin{pmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

The convolution result is denoted by $s = W * x$ and the elements by $s_{i,j}$. Indices start at 1. Values inserted for padding are zeros.

- Assuming stride $S = 1$ and no padding: compute the top right entry $s_{1,5}$. $s_{1,5} =$ _____
- Assuming stride $S = 2$ and symmetric padding size $P = 2$, the output matrix has dimension $D \times D$. Compute D. $D =$ _____
- Assuming symmetric zero-padding size $P = 1$, window-size $W = 3$, and stride $S = 3$, compute the top right entry $y_{1,3}$ after applying 2-max pooling to the input matrix x. $y_{1,3} =$ _____

Question 30: Convolutional Neural Networks

Which statements about CNNs are true? Multiple answers are possible.

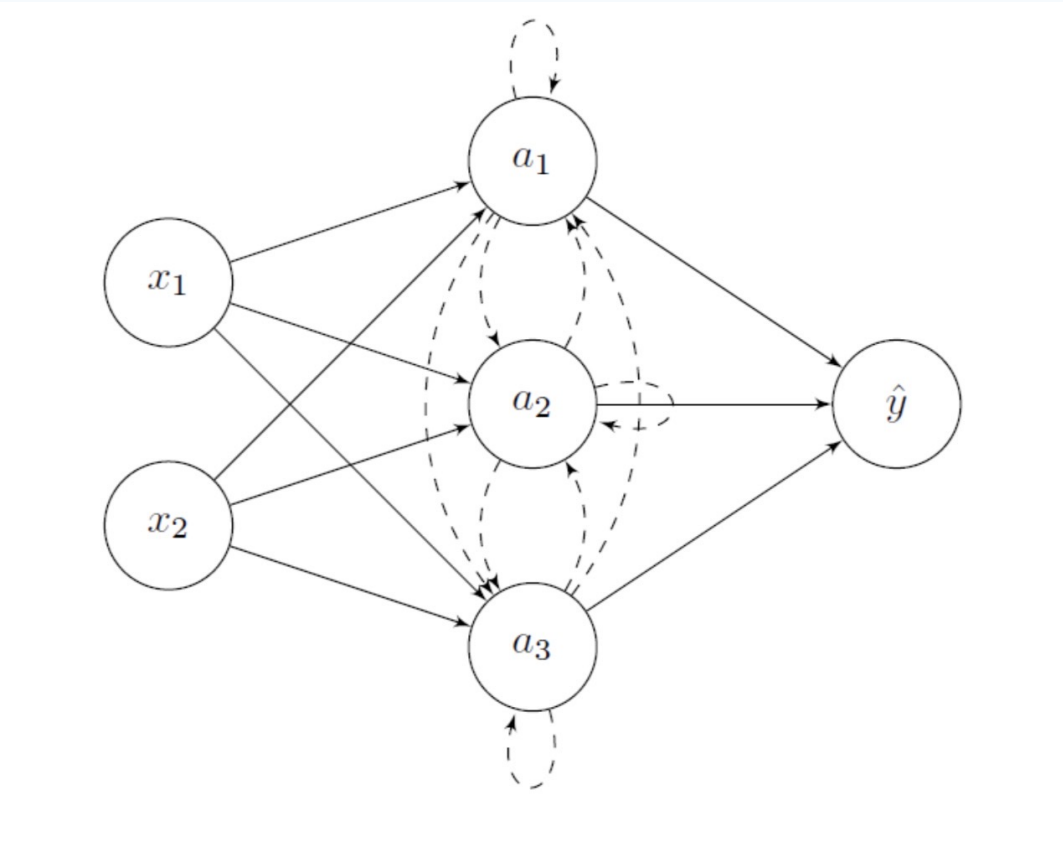
- ☐ In CNNs, the kernel weight matrices must all be different for different positions in the image.
- ☐ Transfer Learning allows using pretrained kernels from other networks which were trained on different data sets.
- ☐ Re-using a kernel weight matrix at multiple positions of an image is called weight sharing.
- ☐ In CNNs, each neuron in a feature map depends on the full input image.
- ☐ The method Deep Dream can be used to improve classification performance by applying it on training data.
- ☐ In Convolutional Layers, a neuron is only connected to a local neighborhood in the

previous layer.

- ☐ Given the feature maps in the last layer, it is possible to revert the operations of the CNN and reconstruct the original input.
- ☐ In CNNs, all kernels need to have the same size.

Question 31: Forward Pass in RNN

Assume the following simple recurrent neural network, where the current preactivation $\hat{p}^{(t)}$ and activation $\hat{y}^{(t)}$ are computed as shown in the figure:



Let $a^{(t-1)} = (3, 0, 0)^\top$, $\hat{y}^{(t-1)} = 3$, and $x^{(t)} = (2, 1)^\top$. Furthermore, let

$$W = \begin{pmatrix} 1 & 3 & 1 \\ 2 & 1 & -1 \end{pmatrix}, \quad R = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 1 & 3 \end{pmatrix}, \quad V = \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}.$$

The activation function of the hidden units is ReLU, and the activation function of the output unit is sigmoid:

$$\sigma(s) = \frac{1}{1 + e^{-s}}.$$

Compute $\hat{p}^{(t)}$ and $\hat{y}^{(t)}$. Give answers rounded to two decimal places.

$\hat{p}^{(t)} =$ _____ $\hat{y}^{(t)} =$ _____

Answer Key

Q1. 0.81

Q2. Accuracy = 0.83; Balanced accuracy = 0.85

Q3. Model variance = 36; Unavoidable error = 16; Bias = 5; Expected prediction error = 77

Q4. 0.78

Q5. 0.91

Q6. $L(\theta_1) = 0.36$; $L(\theta_2) = 0.33$; better: $i = 1$

Q7. $i_1 = 4,\ i_2 = 5,\ i_3 = 2,\ i_4 = 3,\ i_5 = 1$

Q8. True; $p = 0.5 \Rightarrow 2.77$; $p = 0.58 \Rightarrow 2.50$

Q9. Blue: 5; Red: 6

Q10. $w_1^\star = 1$; $w_2^\star = 3$

Q11. 0.14

Q12. -1

Q13. **b, e, h**

Q14. **a, b, c**

Q15. **c, e, f, g**

Q16. **d, e, f, h**

Q17. **b, d, f**

Q18. -1

Q19. -1

Q20. **(4,1)**

Q21. **d, e, f**

Q22. **b, c, e**

Q23. **c, d**

Q24. 0.95

Q25. 0.62

Q26. **c, d, h**

Q27. $\hat{y} = -4$; $\delta_2 = 5$; $\frac{\partial L}{\partial w_{21}} = 10$; $w_{21}^{\text{new}} = 1.9$; $w_{31}^{\text{new}} = -1$

Q28. **c, f, h**

Q29. $s_{1,5} = -2$; $D = 5$; $y_{1,3} = 2$

Q30. **b, c, f, g**

Q31. $\hat{p}^{(t)} = -4$; $\hat{y}^{(t)} = 0.02$